A2: Generative Artifact Proposal- The Sound of a Name

1. Idea

A name is one of the first things we are given, and it becomes a part of our identity. For this project, I want to turn a name into music and explore how numbers can be used to create something personal.

Each letter is given a number based on its position in the alphabet (a = 1, b = 2, and so on). These numbers are then used to generate patterns using the Fibonacci sequence and modular arithmetic. Each letter creates a different repeating pattern, which is used to build parts of a song, such as the chords, bass, melody and drums.

I want to see if a system built using the same set of rules can create music that feels different for every name. Even though the maths stays the same, changing the name should change how the music sounds and how the final artwork looks.

The music will also be represented as a circular, vinyl record like design. This allows the song to be experienced in two ways: by listening to it and by looking at its visual pattern.

2. How the system works

What stays the same for every name

- The music uses a 4/4 time signature, with each beat divided into four smaller steps.
- The same sounds are used for every song: kick, snare, hi-hat, chords, bass and lead melody.
- The basic drum pattern stays the same, with the kick on every beat, the snare on beats 2 and 4, and the hi-hat playing between the beats.

What changes based on the name

- **Letters become numbers:** Each letter is converted into its position in the alphabet.
- **Numbers create patterns:** Each number is used to generate a Fibonacci sequence using modular arithmetic. The sequence repeats in a loop, creating a different pattern for each letter.
- **The name sets the tempo and key:** Adding all the letter numbers gives a total. This total determines the speed of the song and its starting note.

- **The first letter chooses the scale:** The first letter determines which musical scale is used.
- **The length of the name affects the sound:** The number of letters determines how much the chord sounds are detuned, creating a slightly unusual, layered sound.
- **Each loop creates music:** Every value in a loop represents a beat. A value of zero creates silence, while other values determine which note is played.
- **Each letter gets its own section:** The length of a letter's loop determines how long its section lasts. The first value of each bar also determines the chord.
- **The bass and melody work together:** The bass plays on the beat, while the lead melody echoes it between the beats, at a higher pitch.
- **The numbers change the drums:** Certain values add extra hi-hats or softer snare hits.
- **The chords change in volume:** Each letter controls a slow rise and fall in the volume of the chords, adding movement to the music.
- **Spaces between words:** A space creates one bar of drums without the other instruments.

What is random?

A random seed adds tiny changes to the timing and volume of the sounds. This means the same name can have slightly different versions of the song, without changing the notes or the main pattern.

The name determines the main structure of the music, while the seed adds small variations to each version.

Techniques used

The project uses modular arithmetic, Fibonacci sequences, loops and conditional statements, seeded randomness, sound synthesis, volume changes and polar coordinates to create the circular artwork.

Worked example: "maya ramesh"

Each letter is converted into a number:

m = 13, a = 1, y = 25, a = 1, r = 18, a = 1, m = 13, e = 5, s = 19, h = 8.

The total is 104, which gives the song a tempo of 104 BPM and determines its starting note. The first letter, m, determines the musical scale.

Each letter then generates its own Fibonacci loop. For example, the loop for m, using modulo 13, begins with the values 0, 1, 1, 2. These values become a rest followed by three notes, creating the beginning of that letter's musical section.

The different loops are combined to create the full song. In this example, the song is around 2 minutes and 23 seconds long.

3. References

I will study these references and add my own observations about how each one influences my project.

- **Nadieh Bremer, Sony Music's *Gold Record* data art:** This project turns music data into unique, circular artworks that look like records. It inspires my idea of representing a song as a printed disc, where the patterns can be explored visually.
- **Brian Eno, *Music for Airports* (1978):** This piece uses repeating musical loops of different lengths that overlap and interact with one another. It inspires my use of repeating Fibonacci patterns to create music that changes over time.
- **Ryan Hedgecock's YouTube video on how musical sound is built from the ground up:** This video explains how sounds are created using waves, envelopes, LFOs and other techniques. I used these ideas to decide how the different instruments in my system would sound.

4. Final form

The final outcome will be a printed disc that represents the music created from a name.

- **Main artwork:** An A2 print (42 × 59.4 cm) with one ring for each letter. The first letter will be in the centre, with the other letters arranged around it. A legend will explain what the different dots and shapes mean. The name will be placed beside the disc without overlapping the rings.
- **Series of prints:** Four A4 prints of different names will be displayed next to each other. This will show how the same system creates different visual patterns for different people.
- **Layered prototype:** I will experiment with printing the rings on tracing paper or acetate. This will allow me to separate the letters and see how the patterns look individually and when layered together.
- **Sound and interaction:** The music will play through a web prototype. A QR code on the printed artwork will link to the website so people can listen to the song represented by the disc.

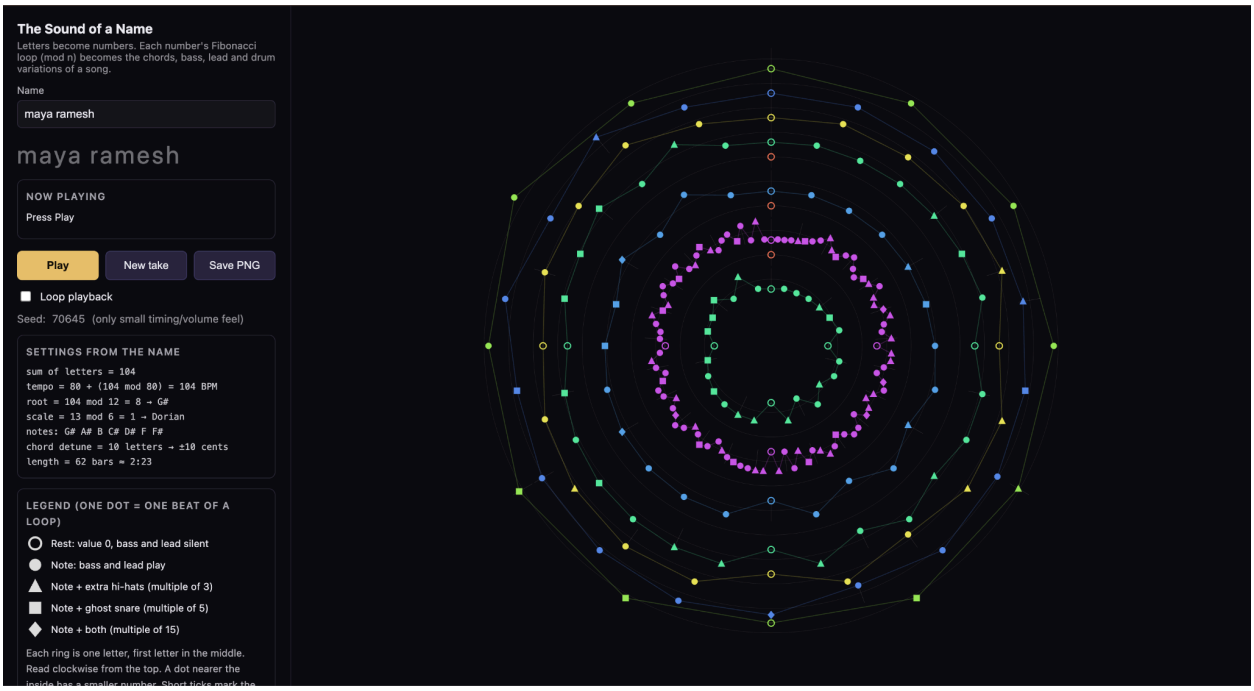
5. Initial sketches and prototypes

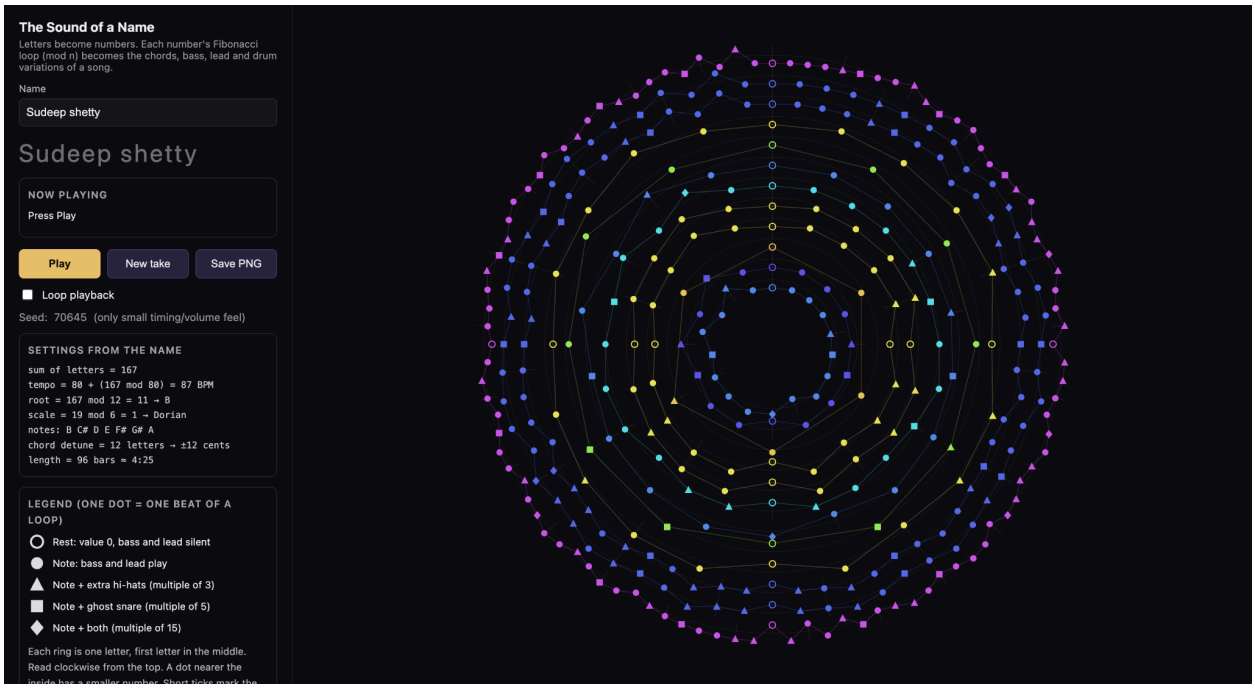
- **Digital prototype:** The HTML file, *name-fibonacci-song.html*, allows users to enter a name, generate a song and see its circular visual representation.
- **Screenshots:** I will save screenshots of the following:
    1. The name "maya ramesh" with the legend visible.

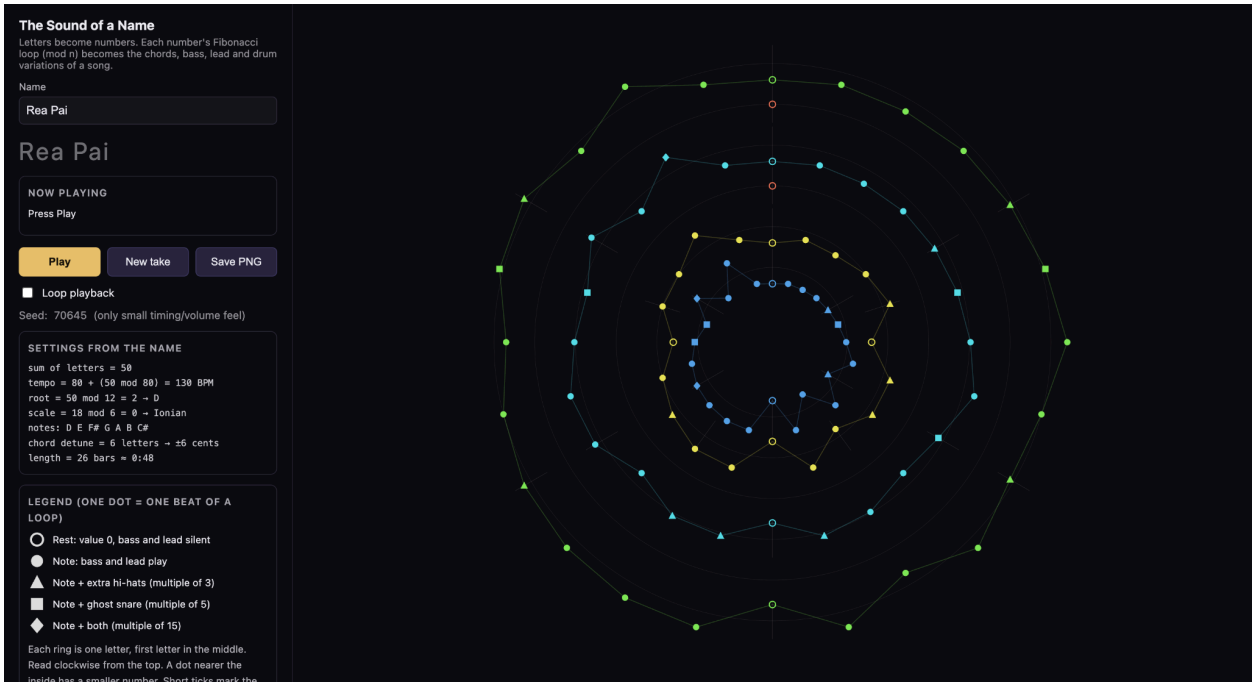
2. Three other names, showing how the tempo, musical key, song length and visual patterns change.

6. Next steps for A3

- Finalise the size and material of the printed artwork.
- Choose four names for the series and generate their corresponding musical and visual outputs.
- Refine the artwork based on the digital and physical prototypes.







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